AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	Liquid Web, Inc, MI -- station KLAN, America/Detroit
Building	OSM 545060574 Liquid Web, Inc
Facade gap	not applicable -- one building, no second facade
Weather record	43,703 real hourly records from KLAN
Plume physics	NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-04-28 (chatter)
Plant limit	21.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 9 of 24 hours with 2 mode changes. That is 2 more than the reactive incumbent, at 0 unsafe hours against its 0. 8 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	9 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent	7 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
	8 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	18.605	21.0	18.330	1.040
01	FREE	yes	-	18.335	21.0	17.780	0.934
02	FREE	yes	-	18.610	21.0	18.330	0.856

03	FREE	yes	-	19.165	21.0	19.440	0.790
04	FREE	yes	-	18.615	21.0	18.890	0.722
05	FREE	yes	-	19.190	21.0	19.440	0.672
06	FREE	yes	-	19.995	21.0	19.440	0.819
07	FREE	yes	-	18.890	21.0	16.670	0.993
08	FREE	yes	-	20.000	21.0	16.110	1.397
09	mechanical	no	dry-bulb	22.220	21.0	17.780	2.035
10	mechanical	no	dry-bulb	22.225	21.0	20.000	1.958
11	mechanical	no	dry-bulb	22.775	21.0	22.220	1.770
12	mechanical	no	dry-bulb	23.055	21.0	22.220	1.510
13	mechanical	no	dry-bulb	21.945	21.0	19.440	1.021
14	mechanical	no	dry-bulb	21.665	21.0	17.220	1.069
15	mechanical	no	dry-bulb	21.665	21.0	18.330	0.867
16	mechanical	yes	switch budget	19.995	21.0	18.330	0.968
17	mechanical	yes	switch budget	18.330	21.0	18.330	0.818
18	mechanical	yes	switch budget	18.055	21.0	17.220	0.973
19	mechanical	yes	switch budget	17.500	21.0	16.670	1.058
20	mechanical	yes	switch budget	16.695	21.0	15.560	1.108
21	mechanical	yes	switch budget	15.275	21.0	13.330	1.416
22	mechanical	yes	switch budget	14.170	21.0	11.670	1.242
23	mechanical	yes	switch budget	13.060	21.0	10.560	0.981

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.605 C, which is 2.395 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.040 C, and the margin is measured, not chosen: 1.040 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.335 C, which is 2.665 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.934 C, and the margin is measured, not chosen: 0.934 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.610 C, which is 2.390 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.856 C, and the margin is measured, not chosen: 0.856 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 19.165 C, which is 1.835 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.790 C, and the margin is measured, not chosen: 0.790 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.615 C, which is 2.385 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.722 C, and the margin is measured, not chosen: 0.722 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 19.190 C, which is 1.810 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.672 C, and the margin is measured, not chosen: 0.672 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 19.995 C, which is 1.005 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.819 C, and the margin is measured, not chosen: 0.819 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.890 C, which is 2.110 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.993 C, and the margin is measured, not chosen: 0.993 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 20.000 C, which is 1.000 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.397 C, and the margin is measured, not chosen: 1.397 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.220 C against a 21.0 C limit, so it fails by 1.220 C. It would take a limit 1.220 C higher, or a bound 1.220 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.225 C against a 21.0 C limit, so it fails by 1.225 C. It would take a limit 1.225 C higher, or a bound 1.225 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.775 C against a 21.0 C limit, so it fails by 1.775 C. It would take a limit 1.775 C higher, or a bound 1.775 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.055 C against a 21.0 C limit, so it fails by 2.055 C. It would take a limit 2.055 C higher, or a bound 2.055 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.945 C against a 21.0 C limit, so it fails by 0.945 C. It would take a limit 0.945 C higher, or a bound 0.945 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.665 C against a 21.0 C limit, so it fails by 0.665 C. It would take a limit 0.665 C higher, or a bound 0.665 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.665 C against a 21.0 C limit, so it fails by 0.665 C. It would take a limit 0.665 C higher, or a bound 0.665 C tighter, to change this hour.

16:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 19.995 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

17:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 18.330 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

18:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 18.055 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

19:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.500 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

20:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.695 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

21:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.275 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

22:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.170 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

23:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.060 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

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